

Startup Intelligence GraphRAG Handbook

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Project root: `/home/ahmad/AI/startup-intelligence-graphrag`

This handbook documents the project from code and real artifacts only. No synthetic results are introduced here.

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Project Snapshot

This repository implements a startup/company intelligence GraphRAG stack over SEC filings.

Ground-truth execution state

- Core run summary exists and is populated:
 - `artifacts/run_summary.json`
- Strict dataset corpus manifest exists:
 - `artifacts/raw/manifest.json`
- Last strict host completion manifest is failed:
 - `artifacts/run_completion_manifest.json`
 - `status: failed`
 - failure message: `Stage 'core_pipeline' failed with exit code 1`

Core run snapshot values (`artifacts/run_summary.json`)

- `dataset_repo : deerfieldgreen/stk-sec-filings`
- `n_filings : 6`
- `n_chunks : 1193`
- `n_eval_queries : 1`
- `n_graph_nodes : 178`
- `n_graph_edges : 602`
- `n_ocr_units : 2`
- `n_vision_units : 2`
- `duration_seconds : 1094.24`

Exact Workflow (Code-Accurate)

This is the real orchestrated workflow from current code.

A) Strict host orchestrator

Entrypoint: `scripts/run_complete_host.py`

Pipeline stages defined in code:

1. preflight checks
 - strict dataset policy + HF access
 - required Ollama model availability
 - adapter prerequisites (`unsloth` , `peft` , `trl` , CUDA)
2. core pipeline stage
 - invokes `scripts/run_full_real_project.py`
3. adapter stage
 - invokes `scripts/run_domain_adapter.py --mode execute --force --required`
4. notebook execution stage
 - invokes `scripts/execute_notebooks.py`
5. test stage
 - invokes `pytest -q`
6. post-check validation and final completion manifest write

B) Core pipeline execution

Entrypoint: `scripts/run_full_real_project.py`

Processing order:

1. build filing corpus (`src/ingest.py`)
2. build evaluation queries (`src/eval_query_builder.py`)
3. chunk corpus (`src/chunking.py`)
4. embed + index chunks (`src/vectorstore.py`)
5. extract entities/relations (`src/extractor.py`)
6. build graph + communities (`src/graph.py`)
7. run retrieval/generation/judge/RAG metrics across techniques
8. build multimodal assets and multimodal techniques
9. write per-technique metrics and samples
10. generate figures and run summary artifacts

C) Optional adapter execution

Entrypoint: `scripts/run_domain_adapter.py`

Modes:

- `placeholder` : seed adapter placeholder artifacts
- `execute` : run Unsloth+TRL training and PEFT evaluation if enabled/forced

Architecture

flowchart TD

```
A[HF SEC Data] --> B[Strict Ingestion Adapter]
B --> C[Chunking]
C --> D[Embeddings + FAISS]
```

```

B --> E[Entity/Relation Extraction]
E --> F[Knowledge Graph]
F --> G[Graph Retrievers]
D --> G
D --> H[Hybrid Sparse+Dense Retriever]
G --> I[Generator]
H --> I
I --> J[Judge + RAG Metrics]
B --> K[Multimodal Asset Builder]
K --> L[HTML/Table Units]
K --> M[OCR Units]
K --> N[Vision Units]
L --> O[Multimodal Retriever]
M --> O
N --> O
O --> I
J --> P[Artifacts + Figures + Notebooks]

```

Techniques and Tutorials

Use these chapters as the zero-to-hero reading order.

1. [00 zero to hero roadmap.md](#)
2. [01 core graphrag_pipeline.md](#)
3. [02 hybrid rag sparse dense.md](#)
4. [03 agentic rag_crag.md](#)
5. [04 multimodal rag.md](#)
6. [05 optional domain adapter unsloth peft trl.md](#)
7. [06 evaluation results and interpretation.md](#)

Real Run Results

Source: `artifacts/run_summary.json`

Retrieval metrics

Technique	Precision@3	Recall@3	F1@3	MRR	NDCG@3
vector_baseline	0.0000	0.0000	0.0000	0.0000	0.0000
graphrag_local	0.3333	0.0256	0.0476	1.0000	0.4693
graphrag_global	0.0000	0.0000	0.0000	0.0000	0.0000
graphrag_hybrid	0.3333	0.0256	0.0476	0.3333	0.2346
hybrid_sparse_dense	0.0000	0.0000	0.0000	0.0000	0.0000
agentic_crag	0.3333	0.0256	0.0476	0.5000	0.2961
multimodal_rag	0.0000	0.0000	0.0000	0.0000	0.0000
multimodal_ocr_rag	0.0000	0.0000	0.0000	0.0000	0.0000

multimodal_vision_rag	0.0000	0.0000	0.0000	0.0000	0.0000
multimodal_unified_v2	0.0000	0.0000	0.0000	0.0000	0.0000

Generation metrics

Technique	EM	BLEU	ROUGE-L	METEOR	BERTScore F1
vector_baseline	-	-	-	-	-
graphrag_local	-	-	-	-	-
graphrag_global	-	-	-	-	-
graphrag_hybrid	0.0000	0.0136	0.1053	0.1303	-0.0076
hybrid_sparse_dense	0.0000	0.0118	0.1000	0.0880	0.0446
agentic_crag	0.0000	0.0087	0.1207	0.1642	0.0312
multimodal_rag	0.0000	0.0054	0.0826	0.0763	-0.0037
multimodal_ocr_rag	0.0000	0.0140	0.1176	0.1698	-0.0092
multimodal_vision_rag	0.0000	0.0072	0.1053	0.0868	-0.0078
multimodal_unified_v2	0.0000	0.0127	0.0881	0.1033	0.0362

RAG metrics

Technique	Faithfulness	Context Precision	Context Recall	Answer Relevancy
vector_baseline	-	-	-	-
graphrag_local	-	-	-	-
graphrag_global	-	-	-	-
graphrag_hybrid	0.9500	0.8500	0.9000	0.9200
hybrid_sparse_dense	0.0000	0.0000	0.0000	0.0000
agentic_crag	0.9500	1.0000	0.9000	1.0000
multimodal_rag	0.0000	0.0000	0.0000	0.0000
multimodal_ocr_rag	0.0000	0.0000	0.0000	0.0000
multimodal_vision_rag	0.0000	0.0000	0.0000	0.0000
multimodal_unified_v2	0.0000	0.0000	0.0000	0.0000

Artifacts and File Guide

Run state

- artifacts/run_summary.json

- `artifacts/run_completion_manifest.json`
- `artifacts/raw/manifest.json`

Per-technique outputs

- `artifacts/eval/*_full_metrics.json`
- `artifacts/retrievals/*_retrieval_samples_placeholder.json`
- `artifacts/generations/*_generation_samples_placeholder.json`

Figures

- `artifacts/figures/retrieval_ndcg_comparison.png`
- `artifacts/figures/generation_rougel_comparison.png`
- `artifacts/figures/judge_overall_comparison.png`
- `artifacts/figures/graph_topology_snapshot.png`

Notebook Guide

Source notebooks (tutorial-authoritative):

- `notebooks/startup_intelligence_graphrag_zero_to_hero.ipynb`
- `notebooks/02_hybrid_sparse_dense_rag_startup_intelligence.ipynb`
- `notebooks/03_multimodal_rag_startup_intelligence.ipynb`
- `notebooks/04_multimodal_rag_ocr_vision_startup_intelligence.ipynb`
- `notebooks/05_optional_domain_adapter_unsloth_pefttrl.ipynb`

Executed notebook artifacts:

- `notebooks/executed/*.ipynb`

Known Gaps and Current Limits

- Current benchmark slice has `n_eval_queries=1` ; this limits generalization strength.
- Adapter stage currently has placeholder result files in available artifacts.
- Latest strict host completion manifest is failed and should be rerun to completion in a future execution cycle.

README Quality References (Internet Research)

The README structure and writing choices were aligned to current public guidance:

- GitHub Docs, About the repository README file:
<https://docs.github.com/en/repositories/managing-your-repositorys-settings-and-features/customizing-your-repository/about-readmes>
- GitHub Docs, Best practices for repositories:
<https://docs.github.com/en/repositories/creating-and-managing-repositories/best-practices-for-repositories>
- Open Source Guides, Starting an Open Source Project (README checklist prompts):
<https://opensource.guide/starting-a-project/>
- Google Style Guide, README files:
<https://google.github.io/styleguide/docguide/READMEs.html>
- Google Documentation Best Practices:
https://google.github.io/styleguide/docguide/best_practices.html